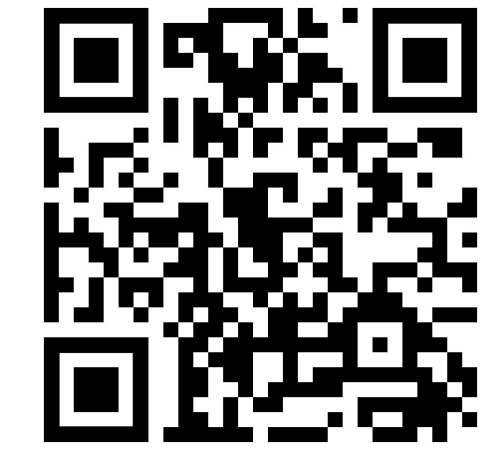


Harvesting Contextuality from the Vacuum

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Basic Idea: Vacuum correlations can generate contextuality in Unruh-DeWitt (UDW) systems using suitable measurements for a single qutrit and qubit-qutrit setup, including gapless systems which cannot harvest entanglement. The qubit-qutrit system also exhibits tunable contextuality-entanglement trade-off relations, allowing for non-classicality to remain in the system if one resource disappears.

Motivating question: are there other resources besides entanglement in quantum fields?

→ Yes, contextuality!

Key question: can an initially non-contextual UDW system state become contextual after a local interaction with the vacuum?

Contextuality Framework

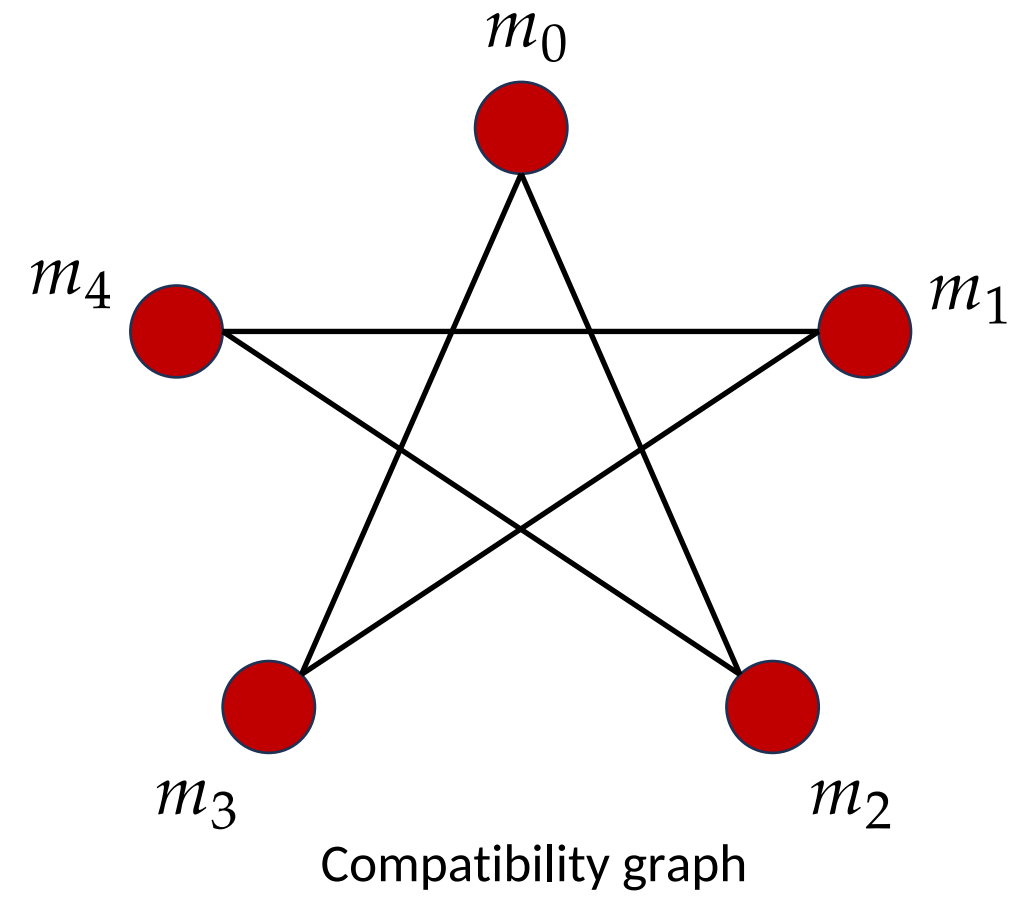
Contextuality asks whether a family of compatible local measurement data admit a single global model independent of context.

Example:

$$X = \{m_0, m_1, m_2, m_3, m_4\}, \quad O = \{-1, 1\},$$

$$\mathcal{C} = \{\{m_0, m_2\}, \{m_0, m_3\}, \{m_1, m_3\}, \{m_1, m_4\}, \{m_2, m_4\}\}$$

	-1, -1	-1, 1	1, -1	1, 1
(m_0, m_2)	0	1/9	2/3	2/9
(m_0, m_3)	0	2/3	1/3	0
(m_1, m_3)	0	1/3	1/3	1/3
(m_1, m_4)	0	1/3	2/3	0
(m_2, m_4)	0	2/3	1/9	2/9



Contextuality generalizes familiar nonclassical resources such as magic [5] and nonlocal entanglement, and is equivalent to Wigner negativity as a measure of non-classicality

Above probability table is called an empirical model [1] and can be written as

$$e = \underbrace{CF(e)}_{\text{Contextual part}} e^{SC} + \underbrace{NCF(e)}_{\text{Non-contextual part}} e^{NC},$$

Contextual part Non-contextual part

→ Measure contextuality with the contextual fraction $CF(e)$

Can phrase as linear program:

$$\begin{aligned} &\text{Find } \mathbf{b} \in \mathbb{R}^n \\ &\text{maximizing } \mathbf{1} \cdot \mathbf{b} \\ &\text{subject to } \mathbf{M}\mathbf{b} \leq \mathbf{v}^e \\ &\text{and } \mathbf{b} \geq \mathbf{0}. \end{aligned}$$

e as a vector:

$$\mathbf{v}^e[(C, s)] = e_C(s).$$

Global assignment matrix:

$$\mathbf{M}[(C, s), g] = \begin{cases} 1 & \text{if } g|_C = s \\ 0 & \text{else.} \end{cases}$$

Single Qutrit Contextuality Harvesting

A single UDW system can harvest non-classicality as magic [3] or contextuality

$$\hat{\rho}_D(t) = \begin{pmatrix} 0 & 0 & -2\bar{W}_{11}^{++} \\ 0 & 2W_{11}^{+-} & 0 \\ -2(\bar{W}^*)_{11}^{++} & 0 & 1 - 2W_{11}^{+-} \end{pmatrix} + \mathcal{O}(\lambda^4)$$

$$\bar{W}_{dd'}^{pq} = \lambda_d \lambda_{d'} W_{\Delta r}(\Lambda_d^p, \Lambda_{d'}^q) \quad W_{dd'}^{pq} = \lambda_d \lambda_{d'} W(\Lambda_d^p, \Lambda_{d'}^q)$$

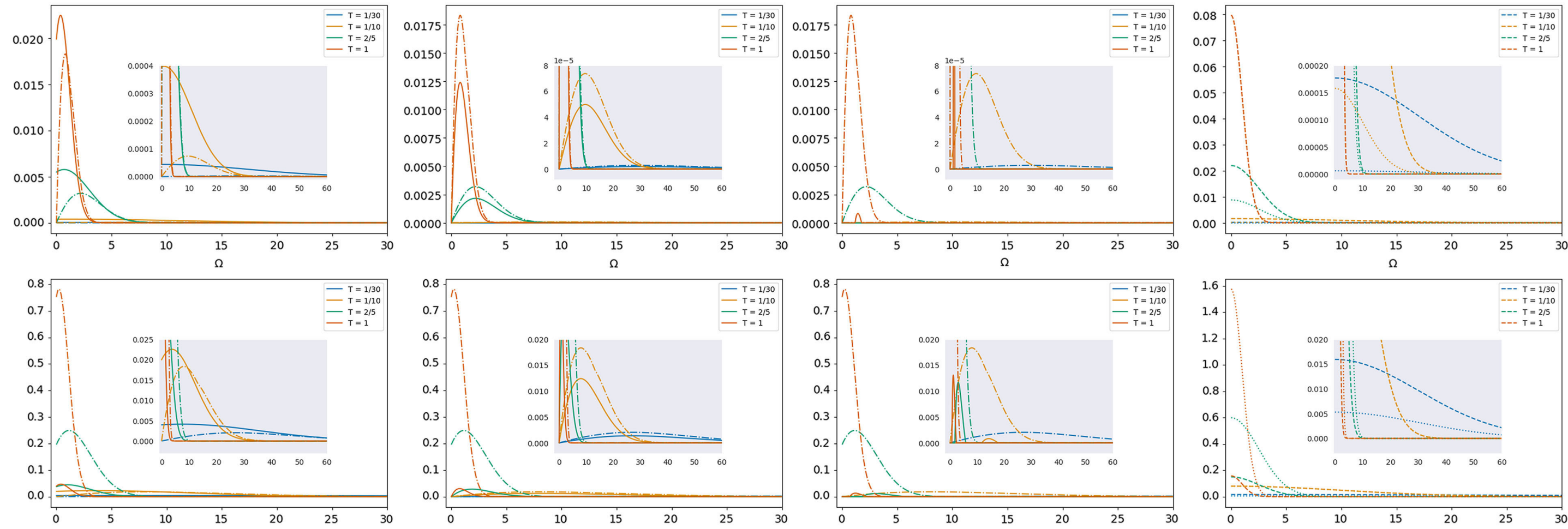


Figure 1: single-qutrit contextuality compared with harvested magic and with the genuine-harvesting diagnostic based on the Hadamard and signalling contributions.

- Sharp temporal profiles preserve genuinely quantum windows
- Longer interactions amplify signalling and can wash out harvested peak
- Smaller spatial smearing strengthens harvested contextuality
- Can tune measurements to alter harvested amount at zero gap, and contextuality can persist after harvested magic has already vanished

Can harvest contextuality and magic for gapless systems depending on the measurement scenario and smearing profile

Key Takeaways

- 1 Single qutrits harvest contextuality from vacuum correlations.
- 2 Gapless systems can harvest contextuality for suitable measurement choices.
- 3 Qubit-qutrit systems can harvest both negativity and local contextuality, with coexistence and trade-offs.
- 4 New genuine-harvesting criteria separate vacuum-resource extraction from field-mediated signaling.

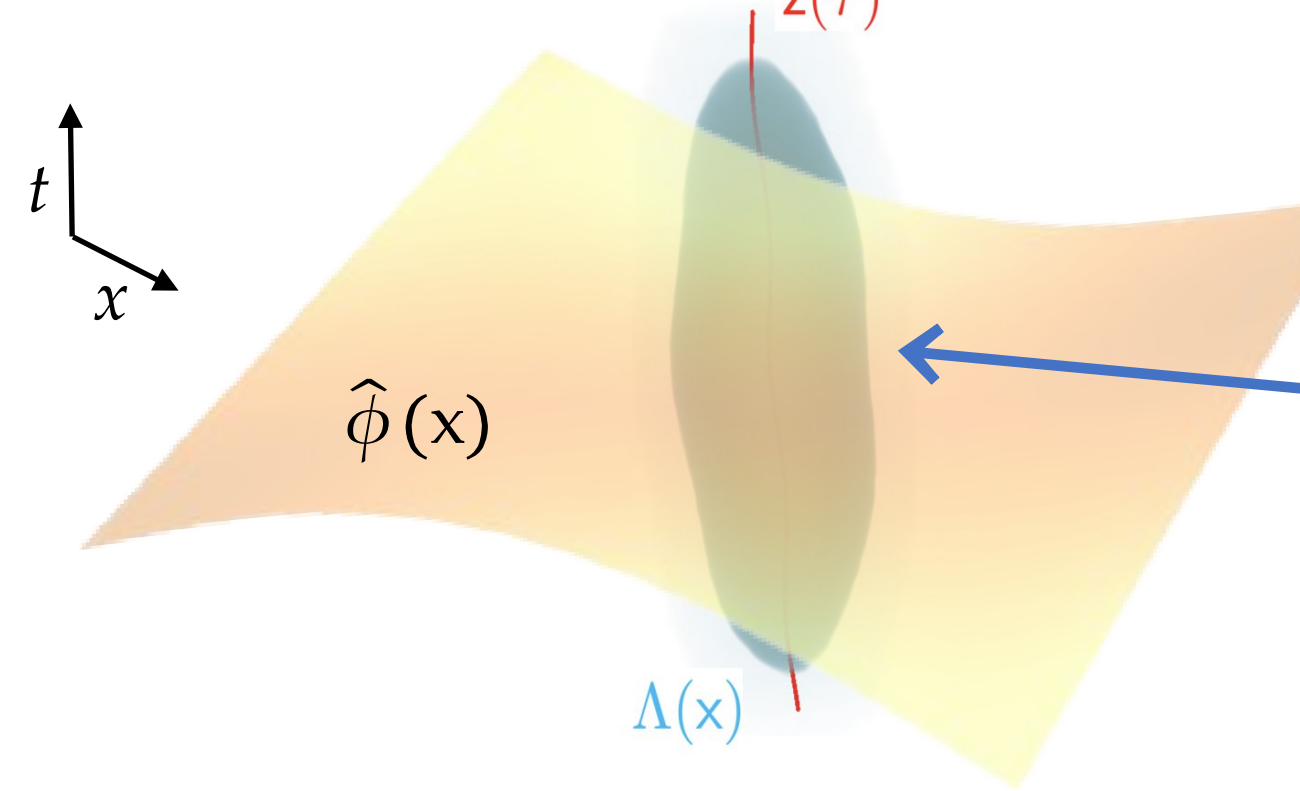
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UDW Harvesting Protocol

Qudit following trajectory through spacetime

$$z^\mu(\tau) = (t(\tau), x(\tau))$$



Internal dynamics prescribed by free Hamiltonian

$$\hat{H}_0 = \sum_d \hat{H}_{0d} = \sum_d \Omega_d (\hat{J}_z^{(d)} + j_d \mathbb{1}_d)$$

Interacts with massless scalar quantum field $\hat{\phi}(x)$ via Hamiltonian density

$$\hat{h}_d(x) = \lambda_d \Lambda_d(x) \hat{\mu}_d(\tau) \otimes \hat{\phi}(x)$$

Interaction smeared out over spacetime region via smearing function $\Lambda_d(x)$

Harvested contextuality measured using

$$\Delta CF(\mathbf{v}^e) = CF(\mathbf{v}_t^e) - CF(\mathbf{v}_0^e).$$

The protocol:

1. Prepare Non-contextual system ground state
2. Couple Local interaction with $\hat{\phi}(x)$
3. Reduce Trace out the field after time evolution
4. Quantify Evaluate $\Delta CF(\mathbf{v}^e)$

Genuine Contextuality Harvesting

Can we call it "harvesting?"

- Vacuum state is highly non-classical [4] → Existing contextuality
- Resource theory of contextuality identifies free operations [6] → Use on field and UDW systems to reduce explanation of acquired contextuality to non-classicality of vacuum

Not all harvested contextuality is due to field

- Hadamard function encodes field state-dependence

Quantify:

$$\left| \frac{\Delta(\Lambda_d^+, \Lambda_{d'}^+)}{H(\Lambda_d^+, \Lambda_{d'}^+)} \right| \ll 1 \quad \text{and} \quad \Delta CF(\mathbf{v}^e) > 0.$$

Genuine harvesting is identified when the Hadamard contribution dominates the signalling term in the relevant peak region

Qubit-Qutrit Contextuality Harvesting

Adding a qubit allows the joint system to harvest negativity while the qutrit still carries local contextuality. The two resource windows need not coincide, so they can compete, align, or separate as the detector profiles and separation are varied.

$$\hat{\rho}_D(t) = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & r_{62}^* \\ 0 & 0 & W_{11}^{+-} & 0 & \sqrt{2}W_{21}^{+-} & 0 \\ 0 & 0 & 0 & 0 & 0 & r_{46} \\ 0 & 0 & \sqrt{2}W_{12}^{+-} & 0 & 2W_{22}^{+-} & 0 \\ 0 & r_{62} & 0 & r_{46}^* & 0 & r_{66} \end{pmatrix} + \mathcal{O}(\lambda^4),$$

$$r_{66} = 1 - W_{11}^{+-} - 2W_{22}^{+-}, \quad r_{62} = -\sqrt{2}(G^*)_{21}^{++},$$

$$r_{46} = -2\bar{W}_{22}^{+-}, \quad G_{dd'}^{pq} = \lambda_d \lambda_{d'} G_F(\Lambda_d^p, \Lambda_{d'}^q)$$

$$\text{Negativity: } \mathcal{N}(\hat{\rho}_D(t)) \approx \left| \min\left(0, -\frac{1}{2}(W_{21}^{+-}W_{12}^{+-} + |r_{46}|^2)\right) + \min(0, -|r_{62}| + 2W_{22}^{+-}) \right| + \mathcal{O}(\lambda^4)$$

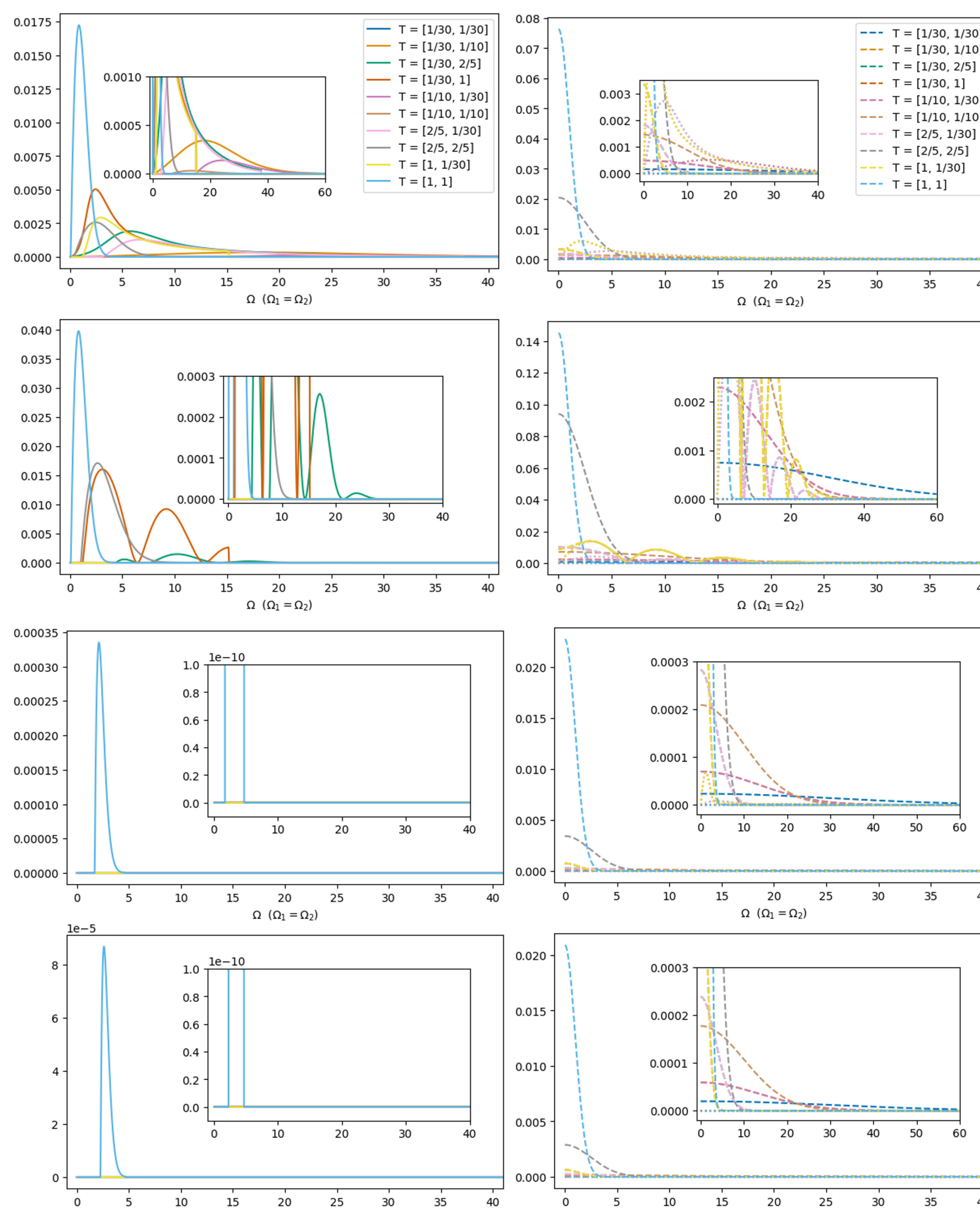


Figure 2: negativity in the qubit-qutrit system as a function of the shared gap, temporal widths, spatial width, and separation.

Harvesting Behaviour

Energy Gap Ω
Both resources fade for large gaps

Spatial Width σ
Smaller smearing generally amplifies harvesting

Temporal Width T
Controls peak alignment and signalling contamination

Separation L
Nonlocal entanglement dies sooner than local contextuality

Coexistence Window
Fix the qutrit in a low-gap, high-contextuality regime and widen the qubit temporal profile to pull the negativity peak toward the same region. Both can coexist inside a controllable overlap region.

Non-classicality remains in the system even at large separation